

# Pharmacist Involvement in Primary Care Improves Hypertensive Patient Clinical Outcomes

Phayom Sookaneknun, Robert ME Richards, Jaratbhan Sanguansermsri, and Chai Teerasut

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**BACKGROUND:** The practice of pharmaceutical care in primary care settings in Thailand is currently not generally accepted.

**OBJECTIVE:** To evaluate the effect of pharmacist involvement in treatment with hypertensive patients in primary care settings.

**METHODS:** The treatment objective was to stabilize the blood pressure (BP) of hypertensive patients in accordance with the Joint National Committee on Prevention, Detection, Evaluation and Treatment of High Blood Pressure guidelines. Patients were randomly assigned to a pharmacist-involved group (treatment) or a group with no pharmacist involvement (control). Pre- and post-test BPs, tablet counts, lifestyle modifications, and pharmacists' recommendations were recorded. The 6-month study was carried out in Mahasarakham University pharmacy and 2 primary care units. Patients were monitored monthly by reviewing their medications and supported by providing pharmaceutical care and counseling.

**RESULTS:** From a total of 235 patients, the treatment group ( $n = 118$ ) had a significant reduction in both systolic (S) and diastolic (D) BP compared with the 117 patients of the control group ( $p = 0.037, 0.027$ , respectively). The 158 patients (76 treatment, 82 control) with BPs  $\geq 140/90$  mm Hg at the beginning of the study showed significant BP reductions ( $p = 0.002$  SBP,  $0.008$  DBP). The proportion of 158 patients whose BP became stabilized was higher in the treatment group ( $p = 0.017$ ). The treatment group showed significantly better adherence ( $p = 0.014$ ) and exercise control ( $p = 0.012$ ) at the end of the study. Physicians accepted 42.72% of medication modifications and 5.34% of the suggestions for additional investigations.

**CONCLUSIONS:** Hypertensive patients who received pharmacist input achieved a significantly greater benefit in BP reduction, BP control, and improvement in adherence rate and lifestyle modification.

**KEY WORDS:** blood pressure, hypertension, pharmaceutical care.

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Hypertension has a marked influence on stroke mortality and other cardiovascular problems.<sup>1</sup> Blood pressure (BP) reductions maintained for a number of years, especially systolic blood pressure (SBP) reductions, have been reported to be associated with 30% and 23% reductions in the number of strokes and coronary events, respectively.<sup>2</sup>

Several investigations have shown the benefit of BP reduction or BP control in reducing the risk of cardiovascular morbidity and mortality.<sup>3-6</sup> Previous studies have shown that introducing pharmaceutical care to hypertensive patients improved the patients' BP control and adherence.<sup>7-9</sup> Pharmaceutical care is defined as "the responsible provision of drug therapy for the purpose of achieving definite outcomes that improve a patient's quality of life."<sup>10</sup> This type of care for hypertensive patients in a primary care setting has not previously been undertaken in Thailand. The study presented here was unique in this respect. The purpose of the study was to evaluate the clinical outcomes for patients being treated for hypertension resulting from in-

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roducing pharmaceutical care through a community pharmacy and in 2 primary care units (PCUs).

## Methods

### STUDY SITE

The study was carried out in Mahasarakham University community pharmacy, near Mahasarakham Hospital in the center of the provincial capital, and in 2 primary care units in Takonyarn and Khamrieng villages. The community pharmacy served a city area, and the primary care units served rural areas located in an area of about 3 km around Mahasarakham University.

The healthcare workers in the PCUs consisted of 2 groups. The first group rotated twice a month and consisted of a physician, a nurse, and a pharmacist from the hospital. These 2 clinic visits provided rural patients treatment equivalent to that provided to city patients by the outpatient department of the hospital. The other group of healthcare workers practicing in the PCUs worked in the PCUs every day. This group, in each PCU, usually consisted of a nurse who had qualified after 4 years of study, another nurse who had a 2-year qualification, and 2 nurses who had 1.5 years of nursing qualification plus a bachelor's degree in public health or a related subject. These 4 individuals provided nursing care and medication for common diseases and referred severe cases to the hospital. A fifth person was involved in a clerical capacity in the day-to-day organization of each PCU. The research pharmacist provided pharmaceutical care in the morning and evening for 3 days each week at the university community pharmacy and in the morning for 4 days each week in the PCUs. The pharmacists who attended the PCUs twice a month and those in the university pharmacy were not involved with caring for patients in the treatment group. Each patient in the treatment group was seen once a month. Occasionally, home visits were made in the afternoons to those few patients who missed their monthly appointment. At the pre- and post-tests, patients were separated to have their BP measured in a different area from the clinic treatment room of the hospital or the PCUs.

### DESIGN

A randomized, pre-test, post-test, controlled group study was carried out between October 2002 and July 2003. The pre-test was undertaken from October through December 2002. We identified an initial sample of 235 patients from Mahasarakham Hospital. A simple randomization technique was used to assign the patients to a treatment group and a control group. BP was measured using a sphygmomanometer.

The work was conducted in compliance with the requirements of the site's Institutional Review Board/Human Subjects Research Committee, and permission was granted by this committee to carry out the study.

### PATIENT SELECTION

The databases from the hospital and 2 PCUs were screened for patients diagnosed as hypertensive. They could be men or women who were >18 years of age and who signed the informed consent form. The eligible patients could also be those who were newly diagnosed during the pre-test period and had the following criteria determined by reviewing their medical records: an average diastolic blood pressure (DBP)  $\geq 90$  mm Hg or an average SBP  $\geq 140$  mm Hg, hypertensive patients with diabetes having an average DBP  $\geq 85$  mm Hg or an average SBP  $\geq 130$  mm Hg, or patients receiving current therapy with antihypertensive drugs (controlled or uncontrolled BP).

Patients were excluded if they had secondary causes of hypertension, which were determined by a review of the patient's medical history and from their other diagnoses, if they were unwilling or unable to return to the PCUs or pharmacy for scheduled appointments, if they planned to move from the area during the study, if they had another family member enrolled in the study, if their SBP was >210 mm Hg or DBP was >115 mm Hg, or if they had a serious complicating disease that was so disabling that BP control was a secondary or minor consideration (terminal cancer, New York Heart Association Class III or IV congestive heart

failure, end-stage renal disease, severe hepatic condition such as cirrhosis, uncontrolled angina pectoris, ventricular arrhythmia, dementia).

### TREATMENT GROUP

Patients in the intervention group were monitored by the research pharmacist for 6 months (January–June 2003). The patients' BP was measured every month as scheduled by the research pharmacist. The measurements were performed in a separate room of the clinic in the morning between 0800 and 1200, as this was the normal time of the clinic. The technique for BP measurement followed established guidelines.<sup>11,12</sup> An adjustable level table top for patients to rest their arms on was used, and different sphygmomanometer cuff sizes were available to ensure the appropriate size for the arm of each patient. Each patient's pharmacy record consisted of the following: demographic data, clinical and therapeutic data, patient behavior, lifestyle, and BP record.

The controlled BP in hypertension without concomitant cardiovascular disease was defined as having a benchmark of  $\leq 135/85$  mm Hg because the conditions of BP measurement were similar to the conditions for self-measurement at home as stated in Joint National Committee on Prevention, Detection, Evaluation and Treatment of High Blood Pressure (JNC-6) guidelines, and all other BP goals also followed JNC-6 guidelines.<sup>12</sup>

The patient consultation consisted of a 30–50 minute face-to-face interview. The research pharmacist assessed the patient's understanding of his or her medications, counseled on the use of their medications, assessed adherence and lifestyle habits, reviewed for adverse events due to drug-related problems, and discussed factors associated with uncontrolled BP and disease state control. This assessment was made from the written patient history and the interview. Drug-related problems were identified, resolved, and prevented. The pharmacist's recommendations for medication regimen changes after detecting drug-related problems were made to physicians, usually by letter from the university pharmacy to the hospital, but also by recording a note in the patient's medical record in the PCUs.

The research pharmacist also adopted a nonpharmacologic approach in providing relevant information and advice for each patient. This covered exercise, fatty diet, salty diet, smoking, alcohol, and weight reduction. The patient's record was updated monthly to include medication provided from the hospital and PCUs.

Educational leaflets and a diary to record lifestyle were presented during the patient's first visit. The leaflets and diary were developed by the research pharmacist. Areas covered were information about hypertension and possible complications, as well as the medicines used to treat hypertension, and a blank table to record notes each day on food eaten, medication taken, alcohol intake, exercise undertaken, and any unusual symptoms.

### CONTROL GROUP

This group had no research pharmacist involvement. Control patients received the traditional service provided by the hospital or the PCUs.

BP measurement was performed at the pre- and post-test periods. The same method of BP measurement was used as in the intervention group.

### DATA ANALYSIS

Sample size was calculated from the pilot test by using the formula of 2 groups of independent subjects. We set  $\alpha$  at 0.05,  $\beta$  at 0.1 (power of 90%), and the effective size of SBP change at 10 mm Hg. The target size of the study sample was thus calculated to be 95 patients. An additional 30% was added to allow for patient drop-outs, making a total of 124 patients per group.

Statistical analysis was performed using SPSS 10.0. BP reduction between groups was determined using the analysis of covariance in the multiple regression model. If interactions were found, only those with clinical significance were included in the model. BP differences between the pre- and post-tests were analyzed by using the paired *t*-test and categorical data by using  $\chi^2$ . Continuous variables in demographic data were analyzed using mean and standard deviations. The significance level was set at  $p < 0.05$ . BP controls, adherence rate controls, and clinical factors

were analyzed by multiple logistic regression using the pre-test in the model.

Adherence rate was calculated by the number of medicines taken divided by the number supplied, multiplied by 100. A rate  $\geq 80$  was considered good adherence;  $< 80$  represented poor adherence.

Intent-to-treat represented the total numbers of patients recruited, that is, a total of 235 patients' BPs were included in the data analysis. Intent-to-treat was used as the basis for inclusion in the study to reduce the bias that would occur if patients who dropped out of the study were not included in the total numbers. Patients who dropped out were determined from the most recent data on the patient history card.

## Results

### PATIENT CHARACTERISTICS

From the sample of 235 patients, 118 were allocated to the treatment group and 117 patients to the group where no pharmacist was involved (control). Eight patients dropped out during the study: 2 in the treatment group at pre-test and 3 at post-test, and a further 3 patients in the control group at the post-test. The baseline patient characteristics at the beginning of the study are shown in Table 1. The results of randomization of hypertensive patients into groups showed no significant differences in demographic vari-

ables and baseline BP between groups, that is, the 2 groups were equal in all variables.

### BLOOD PRESSURE CONTROL

At the beginning of the study, only 27 of 118 patients in the treatment group had both SBP and DBP controlled. This was not significantly different from the number in the control group, where 21 of 117 patients had their BPs controlled ( $p = 0.349$ ). At the end of the study, BP was controlled in 78 patients in the treatment group and 67 in the control group. This represented a significant improvement in BP control in both groups compared with the pre-test results ( $p < 0.001$ ). However, after 6 months, the proportion of patients with BP control was not significantly different between the groups ( $p = 0.061$ ).

The data were also analyzed after excluding patients whose BP was controlled at values  $< 140/90$  mm Hg at the pre-test. This left 158 patients who had elevated BP at the pre-test. Of these, 76 were in the treatment group and 82 in the control group. At the end of the study, 46 patients in the treatment group had both SBP and DBP controlled. This compared with 34 patients in the control group, resulting in a significant difference between the groups ( $p = 0.017$ ).

### BLOOD PRESSURE DIFFERENCE

When comparing BPs at the 6-month follow-up visit with baseline, both the treatment and control groups showed significant decreases in SBP and DBP ( $p < 0.001$ ) in both groups. These results are shown in Table 2.

Table 2 also shows the results for the total number of 235 patients and for the 158 patients who had higher BPs of at least 140/90 mm Hg at the beginning of the study. The results for the SBPs of the 235 patients show that, when the baseline SBP, the groups, and the interaction between these 2 variables were controlled, the treatment group experienced a

**Table 1.** Homogeneity of Demographic Variables Between Groups at Baseline

| Variable                                              | Treatment Group<br>(n = 118) | Control Group<br>(n = 117) | p<br>Value |
|-------------------------------------------------------|------------------------------|----------------------------|------------|
| Gender                                                |                              |                            | 0.224      |
| men                                                   | 42                           | 33                         |            |
| women                                                 | 76                           | 84                         |            |
| Age, y (mean $\pm$ SD)                                | 63.20 $\pm$ 9.33             | 63.23 $\pm$ 9.25           | 0.982      |
| Disease (n)                                           |                              |                            | 0.474      |
| hypertension                                          | 57                           | 54                         |            |
| hypertension with diabetes                            | 39                           | 45                         |            |
| hypertension with target<br>organ damage <sup>a</sup> | 13                           | 7                          |            |
| hypertension with diabetes<br>and target organ damage | 9                            | 11                         |            |

<sup>a</sup>Previous stroke, myocardial infarction, left ventricular hypertrophy, angina, congestive heart failure, transient ischemic attack, renal failure.

**Table 2.** Mean Blood Pressures and Paired Differences in Total Group vs Patients Hypertensive at Baseline<sup>a</sup>

| BP Variable <sup>b</sup><br>(mm Hg) | Total Group (mean $\pm$ SD)         |                                        |            | Pts. Hypertensive at Baseline (mean $\pm$ SD) |                                       |            |
|-------------------------------------|-------------------------------------|----------------------------------------|------------|-----------------------------------------------|---------------------------------------|------------|
|                                     | Pharmacist<br>Involved<br>(n = 118) | No Pharmacist<br>Involved<br>(n = 117) | p<br>Value | Pharmacist<br>Involved<br>(n = 76)            | No Pharmacist<br>Involved<br>(n = 82) | p<br>Value |
| Pre-test between groups             |                                     |                                        |            |                                               |                                       |            |
| systolic                            | 144.76 $\pm$ 19.69                  | 142.41 $\pm$ 19.81                     | 0.600      | 155.19 $\pm$ 15.51                            | 152.19 $\pm$ 16.17                    | 0.235      |
| diastolic                           | 85.72 $\pm$ 13.56                   | 85.96 $\pm$ 12.94                      | 0.889      | 90.47 $\pm$ 13.83                             | 89.73 $\pm$ 12.96                     | 0.731      |
| Post-test between groups            |                                     |                                        |            |                                               |                                       |            |
| systolic                            | 121.47 $\pm$ 14.90                  | 124.77 $\pm$ 17.97                     | 0.037      | 124.16 $\pm$ 14.23                            | 130.36 $\pm$ 16.83                    | 0.002      |
| diastolic                           | 71.55 $\pm$ 10.80                   | 74.23 $\pm$ 11.87                      | 0.027      | 73.08 $\pm$ 10.68                             | 76.52 $\pm$ 12.35                     | 0.008      |
| Paired differences within groups    |                                     |                                        |            |                                               |                                       |            |
| systolic                            | 23.29 $\pm$ 19.10                   | 18.64 $\pm$ 17.67                      | $< 0.001$  | 26.26 $\pm$ 18.14                             | 21.83 $\pm$ 17.84                     | $< 0.001$  |
| diastolic                           | 14.18 $\pm$ 11.20                   | 11.73 $\pm$ 10.08                      | $< 0.001$  | 15.22 $\pm$ 10.95                             | 13.22 $\pm$ 10.37                     | $< 0.001$  |

<sup>a</sup>BP  $\geq 140/90$  mm Hg.

<sup>b</sup>Intent-to-treat.

significantly greater decrease in SBP at the 6-month follow-up than the control group. When the baseline DBP was used as the covariate, the treatment group showed a significantly greater decrease in DBP than the control group. In the population of 158 patients, the treatment group also showed a more significant decrease in both SBP and DBP. No significant interaction between baseline SBP or baseline DBP and patient group was found.

### RESULTS OF PHARMACIST'S INTERVENTION

Table 3 shows the response to the pharmacist's recommendations to modify the medication. As the result of the pharmacist's letters from the community pharmacy, more than half of the suggestions to change the patient's medication were accepted. In addition, 45 of 130 recommendations made by the pharmacist in the patient's notes in the PCUs and 11 recommendations for more investigations to be undertaken were accepted. Thus, a total of 99 of 206 (48.06%) recommendations were accepted.

As shown in Table 4, at the pre-test, 58 patients in the treatment group and 61 in the control group had good adherence. At the post-test, 70 patients in the treatment group and 60 in the control group were considered adherent. There was an interaction between patient group and adherence rate control at the pre-test.

After the 6-month follow-up, participation in regular exercise showed a significant difference between the groups. The proportion of patients who participated in regular exercise was higher in the treatment group (65 of 114) than in the control group (46 of 117;  $p = 0.012$ ). No interaction was found. The rest of the lifestyle factors did not show any significant difference between groups.

### Discussion

The study was conducted during the period when the JNC-6 guidelines were currently used, and we followed those guidelines for the classification, the goal, and the protocol of treatment. But we also used a benchmark of <130/80 mm Hg in diabetic patients for the target goal of treatment as recommended in JNC-7.<sup>13</sup> We focused primarily on SBP because studies have shown that SBP is more valuable in predicting the risk of cardiovascular disease than DBP, especially in middle-aged, diabetic, and older patients.<sup>2,14,15</sup>

This study indicated that some patients maintained a changed behavior of jogging every morning for exercise as the result of advice from the pharmacist.

For the BP control in the sample of 235 patients (Table 2), the treatment group showed significant control only of the SBP. For the sample of 158 patients with elevated BP, the treatment group showed significantly more control both in SBP and DBP. This may be because the therapeutic goal for patients with controlled BP was to maintain the BP at the same controlled level, whereas for hypertensive patients, the goal was both to reduce BP and to obtain better control.

Both SBP and DBP showed significant differences from the baseline for both groups (Table 2). We did not expect that the BP of the control group would decrease as much as it did, since previous studies showed a significant decrease only in the treatment group.<sup>16,17</sup> During the time of our study, however, there was a major campaign by Mahasarakham Hospital called "Good Heart and Good Health," and many related activities were started, such as group discussions for hypertensive patients at the hospital, visits to special seminars with patients from other areas, and activities encouraging exercise in each community. Another possible weakness of this study resulted from an administrative procedure. This involved a label on the patients' notes to identify those who were included in the study; this may have caused physicians and other healthcare providers to monitor these hypertensive patients more closely. Having the research pharmacist as the only person measuring BPs was considered an advantage because it reduced the variation that might have occurred if more people were involved. Despite the above confounding influences on our study, the results show that significant benefits resulted from the involvement of a pharmacist with hypertensive patients. BP was better controlled and greater reductions were obtained.

It was interesting to find that 48% of the pharmacists' recommendations were accepted by the physicians (Table 3) since some physicians had misgivings at the beginning of the study as to the appropriateness of pharmacists being involved in patient care in the primary care setting.

Pharmacist involvement in providing care for hypertensive patients led to improvement in adherence (Table 4), supporting the effectiveness of such services as shown by Taylor et al.<sup>9</sup> It must be acknowledged that other studies of the effect of pharmacist intervention on patient adherence were not as positive; however, their protocols were less rigorous. One study used patient self-reports to measure

**Table 3.** Response of Physicians to the Pharmacist's Recommendations on Treatment Modifications

| Recommendation Tool | Total Recommendations, n (%) | Recommendations Accepted, n (%) |                                  | Recommendations Not Accepted, n (%) | Recommendations Not Seen, n (%) |
|---------------------|------------------------------|---------------------------------|----------------------------------|-------------------------------------|---------------------------------|
|                     |                              | Modification                    | More Investigations <sup>a</sup> |                                     |                                 |
| Letters             | 76                           | 43                              | 6                                | 17                                  | 10                              |
| Notes               | 130                          | 45                              | 5                                | 69                                  | 11                              |
| TOTAL               | 206 (100)                    | 88 (42.72)                      | 11 (5.34)                        | 86 (41.75)                          | 21 (10.19)                      |

<sup>a</sup>Recommendations related to laboratory tests, such as renal function or lipid profile. Patients whose symptoms indicated they were at risk were also referred.

**Table 4.** Patient Adherence Rates of the Treatment and Control Groups

| Group                  | Adherence, n (%) |             | p Value |
|------------------------|------------------|-------------|---------|
|                        | Good (≥80%)      | Poor (<80%) |         |
| Pre-test <sup>a</sup>  |                  |             | 0.534   |
| treatment (n = 112)    | 58 (51.33)       | 54 (48.21)  |         |
| control (n = 109)      | 61 (56.48)       | 48 (44.04)  |         |
| Post-test <sup>b</sup> |                  |             | 0.014   |
| treatment (n = 110)    | 70 (63.64)       | 40 (36.04)  |         |
| control (n = 108)      | 60 (55.56)       | 48 (44.86)  |         |

<sup>a</sup>Fourteen patients were missing, 8 of whom dropped out and 6 of whom did not provide data. These were not included in this analysis at the pre-test.

<sup>b</sup>By the end of the study, a total of 17 patients were missing. The extra 3 missing patients were not included in this analysis.

adherence<sup>9</sup> and the other used prescription refill data.<sup>18</sup> In our study, we counted patient medications at one-month intervals at the times of the pre- and post-tests.

## Summary

Our results indicate that hypertensive patients who received regular pharmaceutical care provided by a pharmacist achieved a significantly greater benefit in BP reduction and control, as well as improvement in their adherence rate. These findings indicate that pharmacists play a useful role in management of hypertension in primary care.

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## EXTRACTO

**ANTECEDENTES:** La práctica de la "atención farmacéutica" en el ámbito de atención primaria actualmente no está aceptada en Tailandia. Por "atención farmacéutica" se entiende "la provisión responsable de terapia con medicamentos con el propósito de alcanzar resultados definidos que mejoren la calidad de vida del paciente." Este tipo de cuidado aplicado a pacientes con hipertensión arterial no había sido empleado anteriormente en Tailandia. Por este motivo, el presente estudio es relevante.

**OBJETIVO:** Evaluar los resultados clínicos resultantes de la participación del farmacéutico en el cuidado de pacientes hipertensos.

**MÉTODOS:** El objetivo del tratamiento farmacológico fue estabilizar la presión arterial de pacientes hipertensos, siguiendo las pautas de tratamiento incluidas en el documento del "Joint National Committee-6" (JNC-6). Los pacientes se asignaron aleatoriamente a uno de los siguientes grupos: grupo con participación del farmacéutico (grupo de tratamiento) y grupo sin participación del farmacéutico (grupo control). Se registraron los siguientes parámetros: presión arterial antes y después de la prueba, recuento de tabletas, modificaciones del estilo de vida, y todas las recomendaciones farmacéuticas. El estudio duró 6 meses y se llevó a cabo en la farmacia de la Universidad de Mahasarakham y en 2 unidades de atención primaria. Los pacientes se monitorizaron mensualmente para revisar la lista de medicamentos prescritos y evaluar la intervención farmacéutica.

**RESULTADOS:** De los 235 pacientes que participaron en este estudio, los 118 del grupo de tratamiento tuvieron una reducción significativa en la presión arterial sistólica ( $p = 0.037$ ) y en la diastólica ( $p = 0.027$ ) comparado con la reducción observada en los 117 pacientes del grupo control. Los 158 pacientes (76 en el grupo de tratamiento y 82 en el grupo control) que tenían una presión arterial  $\geq 140/90$  mmHg al comienzo del estudio mostraron reducciones significativas en la presión

arterial ( $p = 0.002$  para la sistólica y  $0.008$  para la diastólica). De estos 158 pacientes, la proporción que alcanzó una estabilización en la presión sanguínea fue mayor en el grupo tratado que en el grupo control ( $p = 0.017$ ). Del mismo modo, el grupo tratado mostró resultados significativamente superiores en cuanto a la adherencia a la terapia ( $p = 0.014$ ) y al control del ejercicio ( $p = 0.012$ ) al final del estudio. Los médicos aceptaron un 42.72% de las modificaciones del tratamiento y un 5.34% de las sugerencias para hacer más pruebas complementarias.

**CONCLUSIONES:** Los pacientes hipertensos que recibieron atención farmacéutica se beneficiaron significativamente en la reducción y control de la presión sanguínea; asimismo, hubo una mayor proporción de mejora de cumplimiento del tratamiento y de modificación de estilo de vida.

Encarnación C Suárez

## RÉSUMÉ

**INTRODUCTION:** La pratique des soins pharmaceutiques en médecine familiale en Thaïlande n'est pas acceptée de façon générale.

**OBJECTIF:** Évaluer l'impact de l'implication du pharmacien auprès des patients hypertendus dans un environnement de médecine familiale.

**MÉTHODOLOGIE:** L'objectif du traitement était de stabiliser la tension artérielle (TA) de patients hypertendus selon les recommandations du JNC-6. Les patients furent assignés aléatoirement au groupe traitement (présence d'un pharmacien) et au groupe contrôle (absence d'un

pharmacien). La TA pre- et post-test, le décompte des comprimés, les modifications de style de vie, et toutes les recommandations des pharmaciens furent comptabilisées. Cette étude de 6 mois fut effectuée à la pharmacie de l'Université de Mahasarakham ainsi que dans 2 unités de médecine familiale. Les patients furent monitorés aux mois en revisant leurs médicaments et furent suivis selon le modèle des soins pharmaceutiques.

**RÉSULTATS:** Des 235 patients, le groupe traitement ( $n = 118$ ) a eu une réduction significative de la TA systolique (TAS) ( $p = 0.037$ ) et diastolique (TAD) ( $p = 0.027$ ) lorsque comparé au groupe contrôle ( $n = 117$ ). Les 158 patients ( $n = 76$  groupe traitement et  $n = 82$  groupe contrôle) ayant une TA  $\geq 140/90$  mm Hg au début de l'étude ont démontré une réduction significative de TA ( $p = 0.002$  TAS et  $p = 0.008$  TAD). La proportion des 158 patients dont la TA est devenue stabilisée est plus élevée dans le groupe traitement ( $p = 0.017$ ). Ce groupe a également démontré une plus grande fidélité au traitement ( $p = 0.014$ ) et un meilleur contrôle d'exercice ( $p = 0.012$ ) à la fin de l'étude. Les médecins ont accepté 42.7% des modifications de médicaments et 5.34% des suggestions relatives à des investigations additionnelles.

**CONCLUSIONS:** Les patients hypertendus ayant reçu un suivi par le pharmacien ont démontré une plus grande réduction de la TA, un meilleur contrôle de la TA ainsi qu'une amélioration de la fidélité au traitement et de style de vie.

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